USN | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

I Semester B. E. Regular / Supplementary Examinations Feb-2024

(Common to AS, CH, IM, ME)

FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator is permitted.
4. Use of mathematics Handbook is permitted. Do not write anything on handbook.

PART-A

1	1.1	A is a 3×4 real matrix and $A \times = B$ is an inconsistent system of equations. The highest possible rank of A is _____.	01
	1.2	The rank of 7×7 identity matrix is _____.	01
	1.3	Let $A = \begin{bmatrix} 2 & -1 \\ -4 & 5 \end{bmatrix}$, the eigenvalues of A^T are _____ and _____.	02
	1.4	Angle between radius vector and tangent to the polar curve $r = ae^{b\theta}$ $(b > 0)$ is _____.	02
	1.5	Find the radius of curvature at any point t on the curve $x = at^2$ and $y = 2at$.	02
	1.6	If $x = u\sin(v)$ and $y = u\cos(v)$ then $\frac{\partial(x,y)}{\partial(u,v)} =$ _____.	02
	1.7	If $z = x\sin(y) + y\cos(x)$ then $\frac{\partial^2 z}{\partial x \partial y} =$ _____.	02
	1.8	Evaluate $\int_0^3 \int_0^x x\,dydx$	02
	1.9	If $(\bar{x}, \bar{y})$ is the center of gravity of a two-dimensional plate that occupies the triangle $0 \le x \le 1, 0 \le y \le x$ with density function xy and mass $\frac{1}{8}$, then $\bar{y} =$ _____.	02
	1.10	The Wronskian of the functions $u = e^{-3x}$ and $v = e^{x/5}$ is _____.	02
	1.11	The general solution of $y'' + y' + 0.25y = 0$ is _____.	02

PART-B

2	a	Apply Gauss Jordon method to solve the system of equations: $x + 2y + z = 3;\ 2x + 3y + 3z = 10;\ 3x - y + 2z = 13$	05
	b	Find the values of a and b for which the system $x + y + z = 3;\ x + 2y + 2z = 6;\ x + ay + 3z = b$ have i) no solution ii) unique solution iii) infinitely many solutions.	05
	c	Find the dominant eigenvalue and the corresponding eigenvector of the matrix $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by Rayleigh power method taking the initial vector as $[1 \quad 1 \quad 1]^T$. Perform five iterations.	06

Scanned with OKEN Scanner

3	a	Obtain the angle of intersection of the polar curves $r = \frac{a\theta}{1+\theta}$ and $r = \frac{a}{1+\theta^2}$.	08
	b	Find the radius of curvature of the curve $y^2x = a^2(a - x)$, where a is a constant, at the point (a, 0).	08
		OR	
4	a	Obtain Maclaurin series for the function $f(x) = \log_e(1 + \sin(x))$ upto fourth degree term.	08
	b	Find the circle of curvature of the curve $xy = 12$ at (3, 4).	08
5	a	Let $u = 3x + 2y - z$, $v = x - 2y + z$ and $w = x(x + 2y - z)$. Find Jacobian of u, v, w with respect to x, y, z. Interpret the result.	08
	b	If $u = f(r, s, t)$ and $r = \frac{x}{y}, s = \frac{y}{z}$ and $t = \frac{z}{x}$, then find the value of $xu_x + yu_y + zu_z$.	08
		OR	
6	a	If $u = \log_e(\tan x + \tan y + \tan z)$, then find the value of $\sin(2x)\,u_x + \sin(2y)\,u_y + \sin(2z)\,u_z$.	08
	b	A space probe in the shape of the ellipsoid $4x^2 + y^2 + 4z^2 = 16$ enters Earth's atmosphere and its surface begins to heat. After one hour, the temperature at the point (x, y, z) on the probe's surface is $T(x, y, z) = 8x^2 + 4yz - 16z + 600$. Find the hottest point on the probe's surface.	08
7	a	Evaluate $\iint_R xy^2\,dxdy$ where R is the triangular region bounded by $y = 0, x = y$ and $x + y = 2$.	08
	b	Evaluate $\int_0^2 \int_0^{\sqrt{4-x^2}} y^2\sqrt{x^2 + y^2}\,dydx$ by transforming to polar coordinates.	08
		OR	
8	a	Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\,dzdydx$.	08
	b	Evaluate $\int_0^3 \int_1^{4-y} (x + y)\,dxdy$, by changing the order of integration.	08
9	a	Solve the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x^2e^{3x} - \sin(2x)$.	08
	b	The current in an *LRC* circuit is governed by the differential equation $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = E(t)$. A circuit in series has an electromotive force given by $E(t) = 300V$, a resistor of 10Ω, an inductor of 0.25H and a capacitor of 0.001F. If the initial current and the initial charge on the capacitor are both zero, determine the charge on the capacitor at any time $t > 0$.	08
		OR	
10	a	Using the method of variation of parameters, solve the differential equation $\frac{d^2y}{dx^2} + y = \sec(x)\tan(x)$.	08
	b	Reduce the differential equation $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = x^3 + \sin(\log_e(x))$, where $x > 0$, to a linear differential equation with constant coefficients and hence solve the same.	08

Scanned with OKEN Scanner

USN | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

I Semester B. E. Examinations May-2023

(Common to AS, CH, IM, ME)

FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator is permitted.
4. Use of mathematics Handbook is permitted. Do not write anything on handbook.

PART-A

1	1.1	The rank of the matrix $A = \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$, where $a = b \neq c$ is _____.	02
	1.2	If two of the eigenvalues of $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ are 3 and 6, then the eigenvalues of A^{-1} are _____.	02
	1.3	The circle $x^2 + y^2 - 8x = 0$ in polar form is _____.	02
	1.4	The coordinates of the center of curvature on the curvey $= e^x$ at a point where curve crosses y-axis are _____.	01
	1.5	The radius of curvature at the point $(-1,1)$ on the curve $y = x^2 - 3x + 1$, given $y_1 = -1$ and $y_2 = 2$ is _____.	02
	1.6	If $u = x^y$ then the first order partial derivatives are $\frac{\partial u}{\partial x}$ = _____ and $\frac{\partial u}{\partial y}$ = _____.	01
	1.7	Given $f_{xx} = 6x, f_{xy} = 12$ and $f_{yy} = 6y$ then the nature of stationary point at $(-1,5)$ is _____.	02
	1.8	Transform the integral $\int_{\theta=0}^{\pi/2} \int_{r=0}^{\infty} e^{-r^2} r \, dr \, d\theta$ to cartesian form.	02
	1.9	Sketch the region of the integration $\int_0^1 \int_{e^x}^{e} \frac{1}{\log_e y} dy \, dx$	02
	1.10	If 3 and $-\frac{1}{2}$ are the roots of an auxiliary equation, then the corresponding differential equation is _____.	02
	1.11	The particular integral of the differential equation $y'' - 4y = 3^x$ is _____.	02

PART-B

2	a	Test for consistency of the given system of equations and solve by Gauss elimination method. $x + y + z = 6$ $x - y + 2z = 5$ $3x + y + z = 8$	06
	b	The temperatures x_1, x_2, x_3 of a metal plate follow the linear equations, $2x_1 + 8x_2 - x_3 = 11$ $5x_1 - x_2 + x_3 = 10$ $-x_1 + x_2 + 4x_3 = 3.$ Compute x_1, x_2 and x_3 using Gauss-Seidel iteration method. Perform three	05

Scanned with OKEN Scanner

	c	Perform five iterations of the Rayleigh's power method to determine the dominant eigenvalue and eigenvector of the matrix. Use $X_0 = [1,1,1]^T$ as the initial approximation. $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$	05
3	a	Determine the angle of intersection of the curves $r = a\log_e\theta$ and $r = \frac{a}{\log_e\theta}$.	08
	b	Determine the radius of curvature of the curve $x = a\left(cost + \log_e\left(tan\frac{t}{2}\right)\right), y = asin\,t.$	08
		OR	
4	a	Obtain the angle between the radius vector and tangent at any point on the curve $r = a(1 + sin\theta)$.	08
	b	Expand $\log_e(1 + e^x)$ in ascending powers of 'x' up to the term containing x^4.	08
5	a	If $\theta = t^n e^{-\frac{r^2}{4t}}$, find the value of n such that $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\theta}{\partial r}\right) = \frac{\partial\theta}{\partial t}$.	08
	b	Determine the points on the sphere $x^2 + y^2 + z^2 = 4$ that are closest to and farthest from the point $(3,1,-1)$.	08
		OR	
6	a	Prove that $y = f(x + at) + g(x - at)$ is the solution of one-dimensional wave equation $\frac{\partial^2 y}{\partial t^2} = a^2\frac{\partial^2 y}{\partial x^2}$, where a is a constant.	08
	b	A rectangular box open at the top is to have a volume of 32 cubic ft. Determine the dimensions of the box requiring least material for its construction.	08
7	a	Evaluate the given integral by changing the order of integration $\int_0^1\int_{\sqrt{x}}^1\sqrt{1+y^3}\,dy\,dx$.	08
	b	Evaluate the given integral by changing the variables to polar coordinates. $\int_{-1}^1\int_{-\sqrt{1-x^2}}^0\cos(x^2+y^2)\,dy\,dx.$	08
		OR	
8	a	Find the area lying inside the circle $r = asin\theta$ and outside the cardioid $r = a(1 + cos\theta)$.	08
	b	Determine the mass of the tetrahedron bounded by the coordinate planes and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. The variable density $\rho = \mu xyz$, where μ is a constant.	08
9	a	Obtain the general solution of the differential equation $y'' - 6y' + 9y = e^{3x}(x^2 + 7x + 5)$.	08
	b	Solve the following differential equation by the method of variation of parameters $\frac{d^2y}{dx^2} + 16y = 32\sec 2x$.	08
		OR	
10	a	Solve the differential equation $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 2y = xlog_e x$.	08
	b	The displacement $x(t)$ of a mass-spring system is governed by $\frac{d^2x}{dt^2} + \lambda\frac{dx}{dt} + kx = f(t)$, where λ and k are force constants, determine $x = x(t)$, under the conditions $x(0) = 0 = x'(0)$, and $\lambda = 4$, $k = 20$, $f(t) = 10cos2t$.	08

Scanned with OKEN Scanner